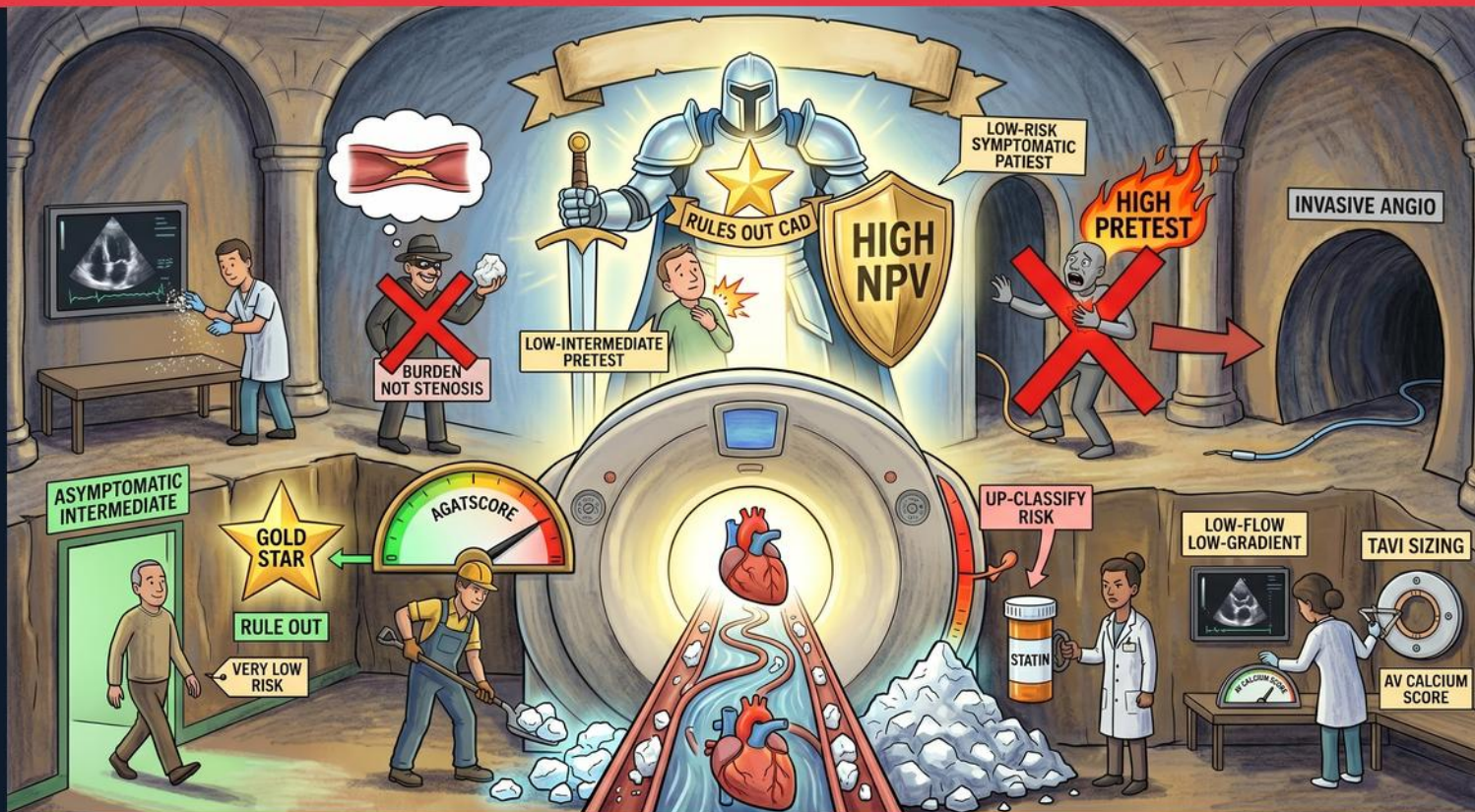


Cardiac Exam — Cardiac CT & Calcium Score





HIGH NPV shield

WHAT YOU SEE

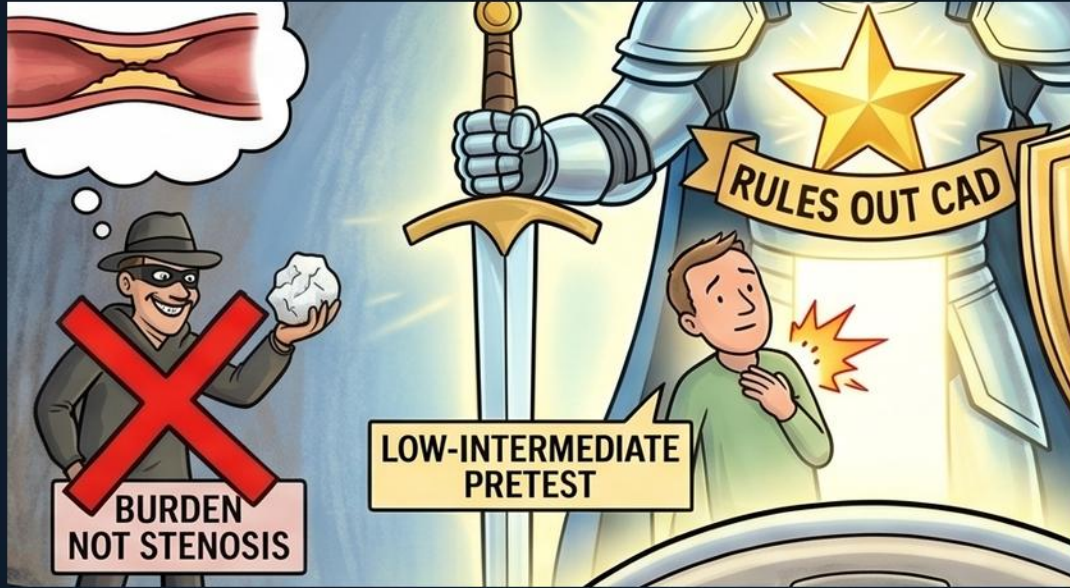
Knight holding a glowing 'HIGH NPV' shield with a sword reading 'RULES OUT CAD'.

WHAT IT ENCODES

Coronary CT angiography (CCTA) has a very high NEGATIVE predictive value: a normal CCTA reliably excludes obstructive coronary artery disease. That makes CCTA the best non-invasive test to RULE OUT CAD. Its strength is exclusion, not confirmation, of disease.

THE BOARD TRAP

Treating CCTA as the test to CONFIRM significant stenosis is wrong — its power is the negative result that rules disease OUT.



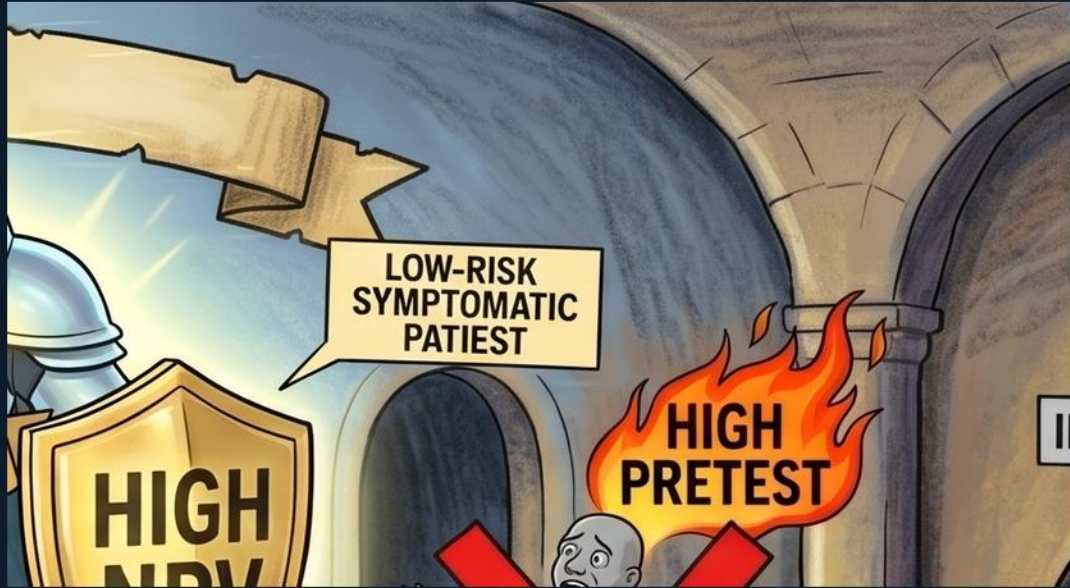
Low-intermediate pretest

WHAT YOU SEE

Card labeled 'LOW-INTERMEDIATE PRETEST' beside a symptomatic patient.

WHAT IT ENCODES

CCTA is best used in SYMPTOMATIC patients with LOW-to-INTERMEDIATE pretest probability of CAD. In that probability band a normal study confidently excludes disease and avoids invasive testing. The pretest probability determines which test is appropriate.



Low-risk symptomatic patient

WHAT YOU SEE

Label 'LOW-RISK SYMPTOMATIC PATIENT' at upper right.

WHAT IT ENCODES

The ideal CCTA candidate is the lower-risk symptomatic patient in whom you want to safely exclude CAD. A negative scan ends the workup; a positive scan can prompt functional or invasive evaluation.



HIGH PRETEST — crossed out

WHAT YOU SEE

Angry red 'HIGH PRETEST' flame figure with a large red X over it.

WHAT IT ENCODES

Do NOT order CCTA in patients with HIGH pretest probability of CAD. High-probability patients are likely to have disease and should proceed to a test that can both diagnose and treat. CCTA in this group adds a step without changing the destination.

THE BOARD TRAP

Ordering CCTA in a high-pretest-probability patient is a classic trap — they belong in the invasive-angiography pathway.



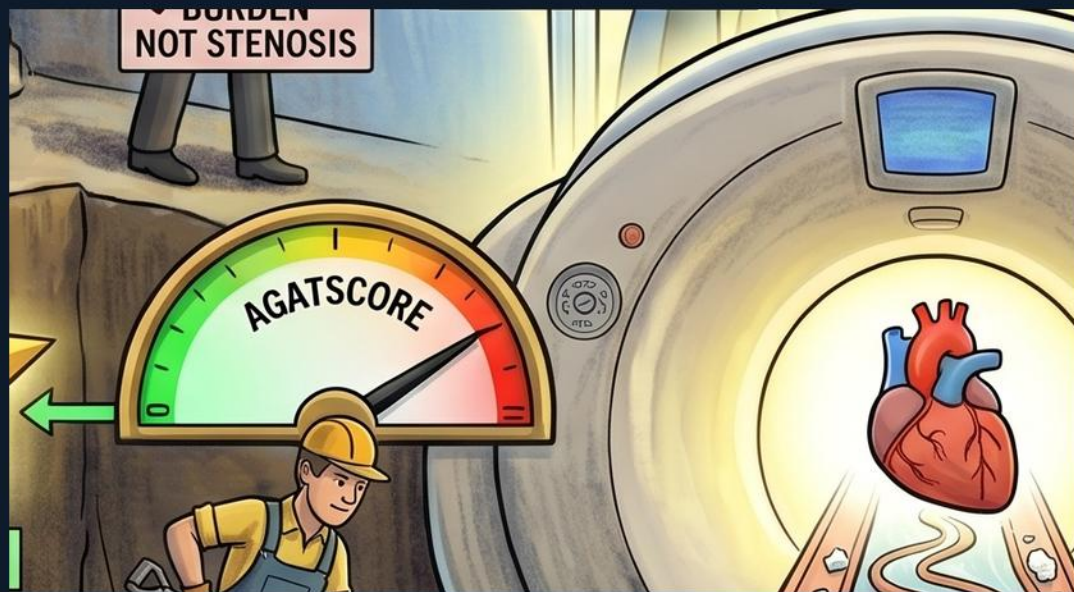
Invasive angiography arrow

WHAT YOU SEE

Tunnel labeled 'INVASIVE ANGIO' with a red arrow pointing into it from the high-pretest figure.

WHAT IT ENCODES

High pretest-probability or high-risk symptomatic patients go directly to INVASIVE coronary angiography. Invasive angiography is both diagnostic and therapeutic, allowing immediate PCI of significant lesions. This is the destination for patients in whom obstructive CAD is already likely.



Agatston calcium gauge

WHAT YOU SEE

A speedometer-style gauge labeled 'AGATSCORE' (Agatston) next to a worker breaking up calcified rocks.

WHAT IT ENCODES

The coronary artery calcium (CAC) score uses the Agatston method to QUANTIFY the burden of CALCIFIED plaque on non-contrast cardiac CT. It is a measure of total atherosclerotic burden, not of any single stenosis. Higher scores indicate more calcified plaque and higher cardiovascular risk.

THE BOARD TRAP

The Agatston score measures calcium BURDEN, not stenosis severity — a high score does not localize or prove an obstructive lesion.



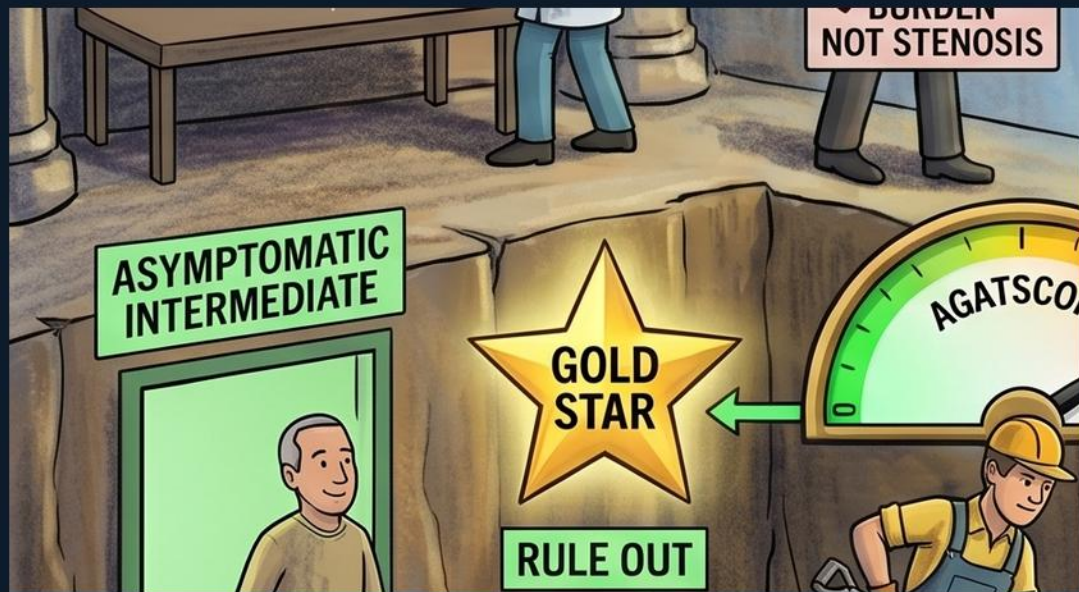
Calcified plaque rubble

WHAT YOU SEE

Pile of white calcified rocks/rubble on the CT table beneath the heart, being mined.

WHAT IT ENCODES

The white rubble represents CALCIFIED atherosclerotic plaque within the coronary arteries. Calcium scoring detects and tallies this calcification. Calcified plaque reflects long-standing atherosclerosis but does not itself indicate the degree of luminal narrowing.



Asymptomatic intermediate door

WHAT YOU SEE

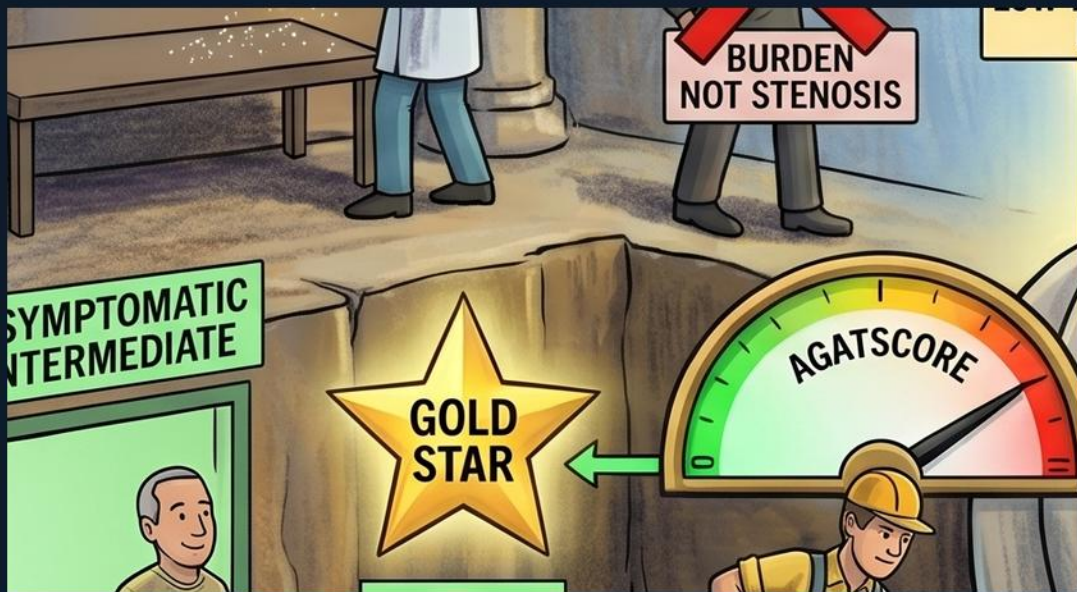
Green doorway labeled 'ASYMPTOMATIC INTERMEDIATE' with a man walking through.

WHAT IT ENCODES

CAC scoring is used for RISK STRATIFICATION in ASYMPTOMATIC patients at INTERMEDIATE cardiovascular risk. It refines risk when the decision to start preventive therapy (e.g., a statin) is uncertain. It is a tool for primary prevention, not for evaluating acute chest pain.

THE BOARD TRAP

CAC scoring is not for symptomatic chest-pain workup — that is CCTA or functional testing; CAC is an asymptomatic risk-refinement tool.



Gold star — score 0

WHAT YOU SEE

Yellow GOLD STAR with a green arrow, labeled 'RULE OUT' and 'VERY LOW RISK'.

WHAT IT ENCODES

A calcium score of ZERO indicates a very low burden of calcified plaque and a very low short-to-intermediate-term cardiovascular risk. A zero score can DE-RISK the patient and may justify withholding or deferring statin therapy. It is one of the strongest negative prognostic markers in prevention.

THE BOARD TRAP

A score of 0 does not exclude soft (non-calcified) plaque, but for risk stratification it strongly de-risks the asymptomatic intermediate patient.



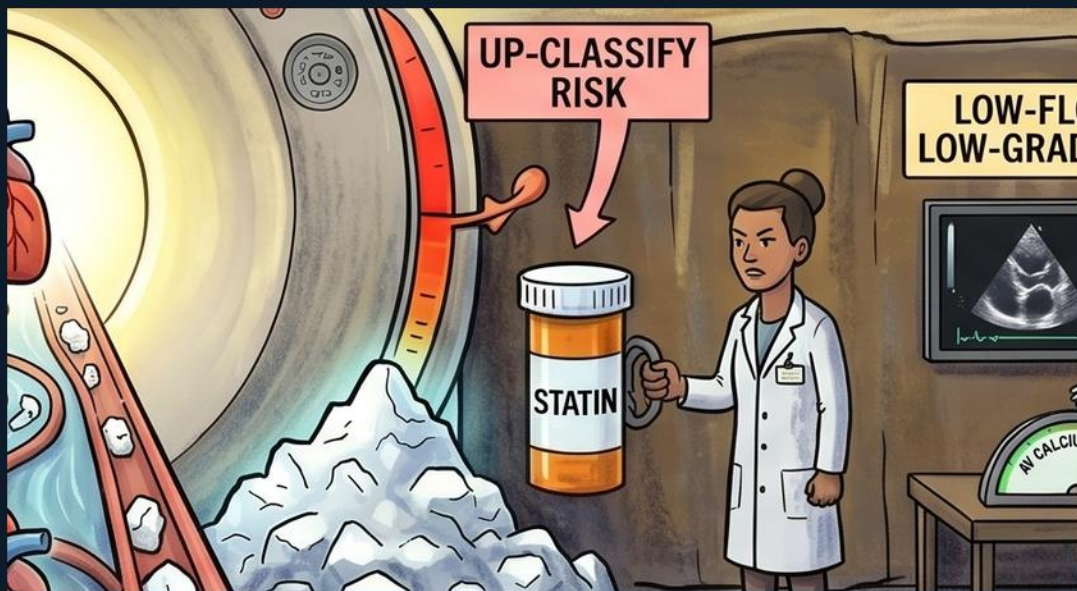
Up-classify risk arrow

WHAT YOU SEE

Pink arrow labeled 'UP-CLASSIFY RISK' pointing down toward a statin bottle.

WHAT IT ENCODES

A HIGH calcium score UP-CLASSIFIES the patient's cardiovascular risk. This reclassification supports initiating or intensifying preventive therapy, particularly statins. The CAC result thus directly changes the management decision in intermediate-risk patients.



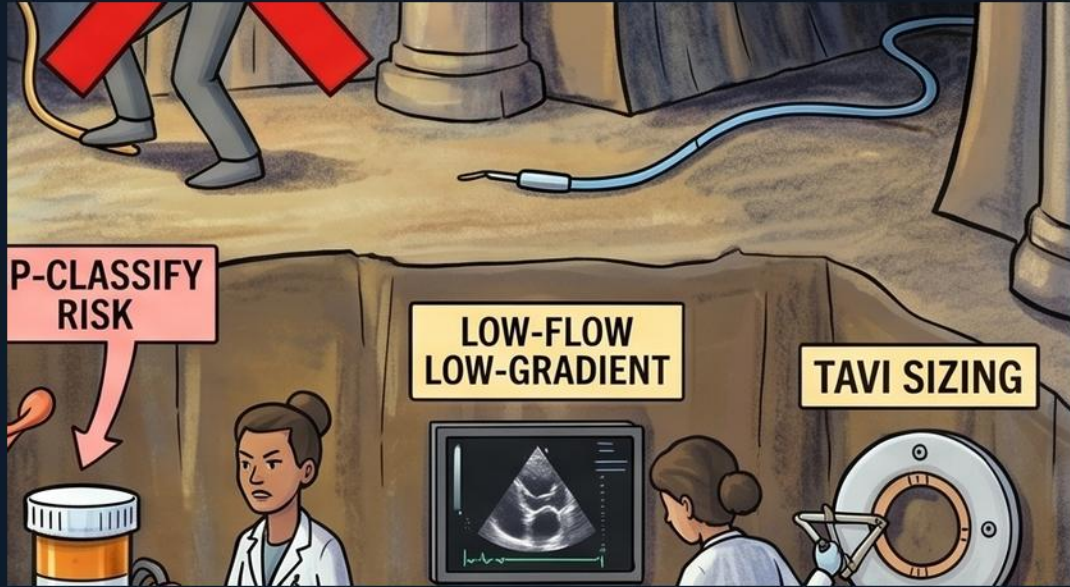
Statin bottle

WHAT YOU SEE

Pill bottle labeled 'STATIN' held by a clinician.

WHAT IT ENCODES

A high CAC score is an indication to START a statin in an asymptomatic intermediate-risk patient, while a zero score may allow withholding it. The score therefore guides primary-prevention pharmacotherapy. This is the practical payoff of risk reclassification.



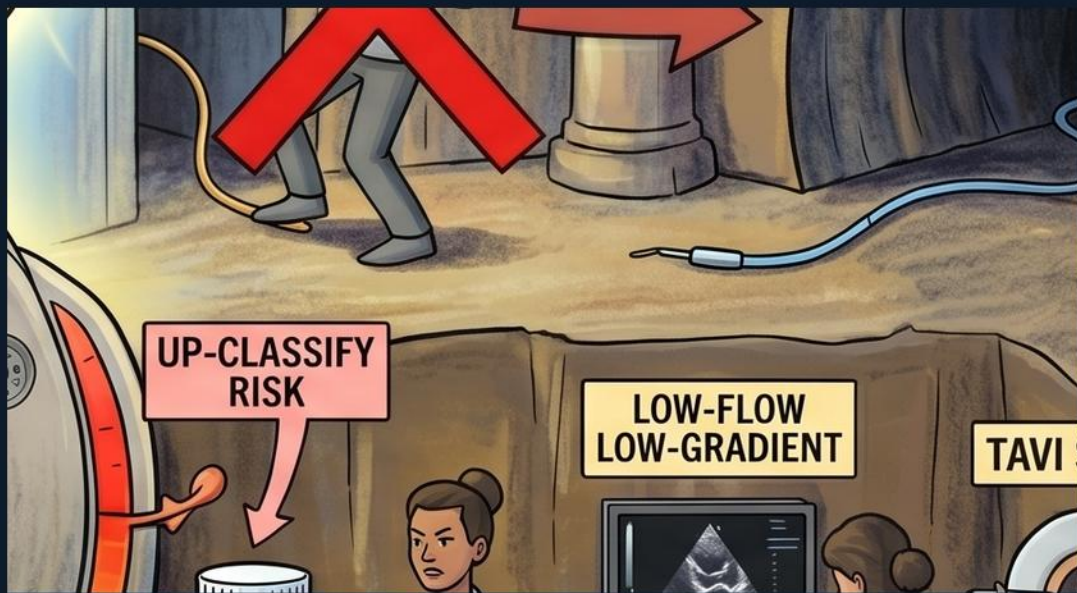
AV calcium score thumbnail

WHAT YOU SEE

Framed CT thumbnail labeled 'AV CALCIUM SCORE' (aortic valve) at lower right.

WHAT IT ENCODES

Aortic valve calcium scoring on CT helps adjudicate aortic stenosis SEVERITY when echocardiography is equivocal. It is especially useful in low-flow, low-gradient AS where Doppler gradients may underestimate true severity. A high valvular calcium score supports severe AS.



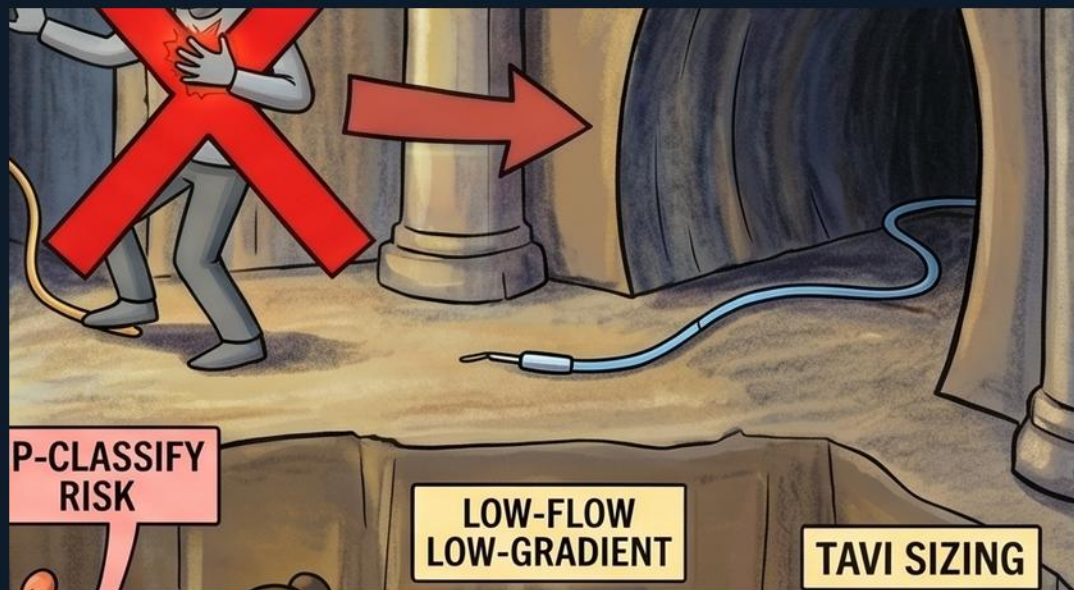
Low-flow low-gradient gauge

WHAT YOU SEE

Gauge labeled 'LOW-FLOW LOW-GRADIENT' beside the AV calcium thumbnail.

WHAT IT ENCODES

In LOW-FLOW, LOW-GRADIENT aortic stenosis, echo gradients can be misleadingly low and severity uncertain. CT aortic valve calcium scoring provides a flow-independent measure to confirm or refute severe AS. This resolves the diagnostic ambiguity that pure Doppler cannot.



TAVI sizing label

WHAT YOU SEE

Label 'TAVI SIZING' near the aortic-valve CT panel.

WHAT IT ENCODES

Cardiac CT is also used to size the aortic annulus and assess access vessels for TAVI (transcatheter aortic valve implantation) planning. Accurate annular measurement determines prosthesis size and reduces paravalvular leak. CT is the standard imaging modality for pre-TAVI workup.



Burden not stenosis

WHAT YOU SEE

Bandit thought-bubble of a calcified plaque with 'BURDEN NOT STENOSIS' and a red X.

WHAT IT ENCODES

A positive calcium score proves the PRESENCE of atherosclerotic burden but does NOT prove obstructive stenosis. Many patients with high calcium scores have no flow-limiting lesion, and some obstructive plaques are non-calcified. Burden and stenosis severity are distinct concepts.

THE BOARD TRAP

Treating any positive calcium score as proof of OBSTRUCTIVE coronary stenosis is a core exam trap — calcium is burden, not a measure of luminal narrowing.

Summary — every symbol

1. HIGH NPV shield

Coronary CT angiography (CCTA) has a very high NEGATIVE predictive value: a normal CCTA reliably excludes obstructive coronary artery disease

3. Low-risk symptomatic patient

The ideal CCTA candidate is the lower-risk symptomatic patient in whom you want to safely exclude CAD

5. Invasive angiography arrow

High pretest-probability or high-risk symptomatic patients go directly to INVASIVE coronary angiography

7. Calcified plaque rubble

The white rubble represents CALCIFIED atherosclerotic plaque within the coronary arteries

9. Gold star — score 0

A calcium score of ZERO indicates a very low burden of calcified plaque and a very low short-to-intermediate-term cardiovascular risk

11. Statin bottle

A high CAC score is an indication to START a statin in an asymptomatic intermediate-risk patient, while a zero score may allow withholding it

13. Low-flow low-gradient gauge

In LOW-FLOW, LOW-GRADIENT aortic stenosis, echo gradients can be misleadingly low and severity uncertain

15. Burden not stenosis

A positive calcium score proves the PRESENCE of atherosclerotic burden but does NOT prove obstructive stenosis

2. Low-intermediate pretest

CCTA is best used in SYMPTOMATIC patients with LOW-to-INTERMEDIATE pretest probability of CAD

4. HIGH PRETEST — crossed out

Do NOT order CCTA in patients with HIGH pretest probability of CAD

6. Agatston calcium gauge

The coronary artery calcium (CAC) score uses the Agatston method to QUANTIFY the burden of CALCIFIED plaque on non-contrast cardiac CT

8. Asymptomatic intermediate door

CAC scoring is used for RISK STRATIFICATION in ASYMPTOMATIC patients at INTERMEDIATE cardiovascular risk

10. Up-classify risk arrow

A HIGH calcium score UP-CLASSIFIES the patient's cardiovascular risk

12. AV calcium score thumbnail

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14. TAVI sizing label

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